Jacopo Gonzini

Milan, Italy | LinkedIn | GitHub

Education

Bachelor's Degree in Economics, Management and Computer Science Expected 2028
Bocconi University, Milan (2025 - Present) | **Weighted Grade Average: 29.4/30**
Interdisciplinary degree combining economics and management with computer science, statistics, and machine learning. Developing a strong foundation in quantitative modelling, data analysis, programming, and the application of computational methods to economic and financial problems.

Relevant Coursework: Advanced Mathematics & Statistics (Linear Algebra, Probability & Statistical Inference), Computer Programming & Numerical Optimization, Macroeconomics, Mathematics & Statistics, Computer Science, Microeconomics.

Liceo Scientifico "D. Bramante", Magenta July 2025
Took part in regional Science Olympiads, ranking among the top 5% of Lombardy students.

Work Experience

Data Science & GenAI Intern, **Accenture**, Milan Jul 2026 - Sep 2026

- Co-designed a Python monitoring and evaluation framework for 14 GenAI agents, targeting approximately 80% lower processing time.
- Built a telemetry pipeline using the Azure Application Insights Python SDK to retrieve and validate Microsoft Foundry traces, storing results in MySQL under privacy constraints.
- Independently built a hybrid Python/Foundry evaluator, cutting token use by 70% versus a full LLM-judge baseline while improving quality and eliminating hallucinated evaluations in testing.

Projects

Research on Spiking Neural Networks · Benchmarks · GitHub Mar 2026 - Present

- Wrote a custom CUDA C kernel from scratch with CuPy RawKernel to simulate Leaky Integrate-and-Fire neuron dynamics, parallelizing state updates across GPU threads.
- Designed a fused execution strategy that moved the full temporal recurrence on-device, achieving 270× speedup over the NumPy CPU backend at 100k neurons on an NVIDIA T4.
- Characterized CPU–GPU crossover and dispatch overhead, showing that GPU acceleration becomes beneficial only beyond sufficient workload size and that stepwise execution is largely bottlenecked by kernel-launch overhead.

Projects (continued)

Empirical Analysis of Global Organ Transplantation · Report · GitHub Mar 2026 - Apr 2026

- Engineered a cross-country analytical dataset by restructuring WHO organ-transplant records, harmonizing external development indicators, and explicitly separating observed zeros, missing values, and structurally unavailable data.
- Investigated missing-data structure and quantified the relationship between human development and transplant activity using Pearson and Spearman correlations across 122 countries.

SPX Volatility Forecasting and Downside Skew · Report · GitHub Aug 2026 - Present

- Built a research pipeline combining daily SPX returns with a 30-day option surface from 2005-2025, measuring downside skew and forecasting realized volatility over the next 22 trading days.
- Implemented leakage-safe, expanding-window HAR and HAR-X models and evaluated 4,865 pseudo-out-of-sample forecasts using bias, MAE, RMSE, skew terciles, and Newey–West inference.
- Found that adding skew to the conditional mean did not improve point forecasts, while skew significantly predicted absolute and squared HAR errors ($p < 0.001$), suggesting value for forecasting uncertainty.

Technical Skills

Programming: Python, R, C++, SQL, CUDA C

ML & Data: PyTorch, TensorFlow, SciPy, NumPy, pandas, CuPy, Matplotlib, Sentence-Transformers, Tidyverse

Stats & Quant: Regression analysis, hypothesis testing, time-series forecasting, Monte Carlo simulation

Tools & Systems: Git, Linux, pytest, Microsoft Foundry, Azure Application Insights, vLLM

Certifications: Text Mining and Sentiment Analysis | Linux for HPC

Activities and Volunteering

Project Division Member, Bocconi AI & Neuroscience Student Association Since Oct 2025
Selected for the project division of Italy's largest student-led AI research lab, contributing to technical projects.

Hackathon Participant, BAINSA accessibility AI Hackathon Sep 2026
Co-developed a 12-hour sound-awareness prototype for Deaf and hard-of-hearing users, consisting of visual smart-glasses cues based on high-accuracy left/right localization and sound classification.

Staff Member, Google Developer Groups Hackathon 2026 May 2026
Supported the organization of a three-day hackathon with 150+ participants.

Other

Languages: Italian (Native), English (Fluent)

Interests: Open-source software, mathematical paradoxes, basketball, chess